

Lecture 4

R_q : Decompositions, Automorphisms, and Embeddings

SEC 595: Encrypted Computing

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Roadmap

Lecture 3 built the ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$ and showed how to multiply in it quickly. That is enough to *state* BGV, BFV and CKKS, but not enough to *run* them. Between the definition of a scheme and a working implementation sit seven pieces of machinery, and this lecture supplies all seven.

They look unrelated and are not. Three are the Chinese Remainder Theorem of Lecture 1 applied to three different rings — the coefficient ring in Section 1, the plaintext ring in Section 5, and over $\mathbb{C}$ the rational ring in Section 7; one is positional notation (Section 3); the rest are the operations those representations make affordable. Each section is exactly three things: the property to be quoted, one paragraph naming what consumes it downstream, and one worked example, complete and checkable. The examples are the explanation.

Throughout, n is a power of two, $R = \mathbb{Z}[X]/(X^n + 1)$ and $R_q = R/qR$; we use the centered representatives and the norm $\|\cdot\|_\infty$ of Lecture 1 and the bound $\|ab\|_\infty \leq n \|a\|_\infty \|b\|_\infty$ of Lecture 3. Worked examples use $n = 4$, or a single coefficient where the identity does not involve n .

1. RNS and the Modulus Chain

Definition 1 (Residue number system, and the modulus chain). Let $Q = q_0 q_1 \cdots q_L$ with the q_i pairwise coprime. The **RNS representation** of $a \in R_Q$ is the tuple $([a]_{q_0}, \dots, [a]_{q_L})$ of its coefficientwise reductions, and

$$R_Q = \mathbb{Z}_Q[X]/(X^n + 1) \cong R_{q_0} \times R_{q_1} \times \cdots \times R_{q_L},$$

an isomorphism for both $+$ and $\times$. Writing $q_\ell = p_0 p_1 \cdots p_\ell$ for $\ell = 0, \dots, L$ gives the **modulus chain**; a ciphertext modulo q_ℓ is said to be at **level** ℓ .

Every practical BGV, BFV and CKKS library stores a ring element this way: an $(L + 1) \times n$ array of machine words, the q_i primes just under 2^{60} , every operation componentwise and no carries between components. BGV, Section 2 takes the chain over verbatim and uses the primes 17, 41, 97 below as its own running example, and CKKS, Section 7 counts levels the same way. Dropping from q_ℓ to $q_{\ell-1}$ deletes one component of the tuple, so L is the multiplicative depth the parameters support and no multiprecision division occurs anywhere.

Example 1 (An RNS round trip, $n = 4$). Take the three primes

$$p_0 = 17, \quad p_1 = 41, \quad p_2 = 97, \quad Q = 17 \cdot 41 \cdot 97 = 67609,$$

each satisfying $p_i \equiv 1 \pmod{8}$, i.e. $2n \mid p_i - 1$, which by Lecture 3 is exactly the condition for a complete length-4 negacyclic NTT to exist in R_{p_i} . By Lecture 1, Section 6.2 the idempotents are

$$e_0 = 63632, \quad e_1 = 52768, \quad e_2 = 18819, \quad e_0 + e_1 + e_2 \equiv 1 \pmod{67609},$$

so a coefficient $a = 12345$ maps to

$$a \mapsto (12345 \bmod 17, 12345 \bmod 41, 12345 \bmod 97) = (3, 4, 26),$$

and is recovered as $3e_0 + 4e_1 + 26e_2 \equiv 12345 \pmod{67609}$. The chain on these primes is $q_2 = 67609$, $q_1 = 697$, $q_0 = 17$.

2. Modulus Switching

Proposition 1 (Modulus switching). *Let ct be a ciphertext modulo q with $ct[SK] = m + tv + qk$ over R , let $q' \mid q$, and let ct' be the integer polynomial closest to $(q'/q)ct$ subject to $ct' \equiv ct \pmod{t}$. If $q \equiv q' \pmod{t}$, then*

$$[ct'[SK]]_{q'} = m + tv' \quad \text{with} \quad \|v'\|_\infty \leq \frac{q'}{q} \|v\|_\infty + \frac{1}{2}(1 + n \|SK\|_\infty) + \frac{\|m\|_\infty}{t},$$

provided the right-hand side is below $q'/(2t)$.

BGV, Section 10 calls this after every multiplication: it is what makes the scheme levelled rather than merely somewhat homomorphic, and each call spends one prime of the chain of Section 1. CKKS, Section 6 runs the same formula $\lfloor (q'/q)ct \rfloor$ with no congruence condition and calls it RESCALE, using it to restore the scale $\Delta^2 \mapsto \Delta$ after a multiplication.

Example 2 ($t = 8$, $q_2 = 67609 \rightarrow q_1 = 697$). Take $q = 67609$, $q' = 697$, so the discarded prime is $p = 97$ and $q/q' = 97$. Track a single coefficient, with secret $SK = 3$; modulus switching is coefficientwise, so nothing is lost.

Let $m = 5$ and $v = 2$, i.e. $ct[SK] = m + tv = 21$ before reduction. With $c_1 = 41234$ we get $c_1 \cdot SK = 123702 \equiv 56093$, hence

$$c_0 = 21 - 56093 \equiv 11537 \pmod{67609}.$$

Check: $11537 + 56093 = 67630 = 67609 + 21$, so $[ct[SK]]_q = 21$ and $21 \bmod 8 = 5 = m$. ✓

Now switch. For each component compute $(q'/q)c$, then move to the nearest integer congruent to c modulo 8:

$$\begin{aligned} c_0 = 11537 : \quad \frac{11537}{97} &= 118.938\dots, \quad c_0 \equiv 1 \pmod{8}, \quad \text{nearest such } c'_0 = 121; \\ c_1 = 41234 : \quad \frac{41234}{97} &= 425.093\dots, \quad c_1 \equiv 2 \pmod{8}, \quad \text{nearest such } c'_1 = 426. \end{aligned}$$

Then $c'_0 + c'_1 \cdot SK = 121 + 1278 = 1399 = 2 \cdot 697 + 5$, so $[ct'[SK]]_{q'} = 5$ and $5 \bmod 8 = 5 = m$. ✓

The noise has gone from $tv = 16$ to $tv' = 0$. Proposition 1 read on this single coefficient (so $n = 1$) predicts $\|v'\|_\infty \leq 2/97 + \frac{1}{2}(1 + 3) + 5/8 \approx 2.65$, which is honoured.

3. Gadget Decomposition

Proposition 2 (Gadget decomposition). *Fix a base $B \geq 2$ and set $\ell = \lceil \log_B q \rceil$. Let $g = (1, B, B^2, \dots, B^{\ell-1})$ be the **gadget vector**, and for $a \in R_q$ let $G^{-1}(a) = (a_0, \dots, a_{\ell-1})$ be the tuple of balanced base- B digits of its centered representative. Then*

$$\langle g, G^{-1}(a) \rangle = \sum_{i=0}^{\ell-1} B^i a_i = a \quad \text{in } R_q, \quad \text{and} \quad \|G^{-1}(a)\|_\infty \leq B/2,$$

the second bound **uniformly in a** .

Whenever a scheme must multiply a ring element by something carrying an error term, it multiplies by the digits instead. That is relinearisation in Section 4 and key switching in Section 6; BGV, Section 9 and BFV, Section 10 publish one evaluation key per digit position on exactly this basis, and GSW, Section 2 is the gadget matrix itself, restating Proposition 2 as *flatness* and citing it as Lecture 4, Proposition 2.

Example 3 (A base- B decomposition). Take $q = 64$, $B = 4$, so $\ell = \lceil \log_4 64 \rceil = 3$ and $g = (1, 4, 16)$. Decompose $a = 45$. The centered representative is $[45]_{64} = -19$, and the balanced base-4 digits are

$$\begin{aligned} -19 &= 4 \cdot (-5) + 1, & a_0 &= 1, \\ -5 &= 4 \cdot (-1) - 1, & a_1 &= -1, & G^{-1}(45) &= (1, -1, -1), \\ -1 &= 4 \cdot 0 - 1, & a_2 &= -1, \end{aligned}$$

so $\langle g, G^{-1}(45) \rangle = 1 - 4 - 16 = -19 \equiv 45 \pmod{64}$. ✓ Here $\|G^{-1}(45)\|_\infty = 1$, inside the bound $B/2 = 2$, while $\|45\|_\infty = 19$. With $B = 2$ instead, $\ell = 6$ and the digits are the ordinary binary expansion $45 = (101101)_2$, i.e. $G^{-1}(45) = (1, 0, 1, 1, 0, 1)$ with $\|\cdot\|_\infty = 1 = B/2$.

4. $R_q[S]$ and Relinearisation

Proposition 3 (Relinearisation). *Identify $ct = (c_0, c_1)$ with $ct[S] = c_0 + c_1 S \in R_q[S]$, so that decryption is evaluation at $S = SK$; a product of two ciphertexts is then the degree-2 polynomial $ct^*[S] = c_1^* + c_2^* S + c_3^* S^2$. Let $(c_{3,0}^*, \dots, c_{3,\ell-1}^*) = G^{-1}(c_3^*)$, let*

$$EK^{(i)} = (-(a_i \cdot SK + e_i) + B^i SK^2, \quad a_i), \quad i = 0, \dots, \ell - 1,$$

and set $\hat{c}_1 = c_1^* + \sum_i EK_1^{(i)} c_{3,i}^*$ and $\hat{c}_2 = c_2^* + \sum_i EK_2^{(i)} c_{3,i}^*$. Then $\hat{ct} = (\hat{c}_1, \hat{c}_2)$ is a degree-1 ciphertext under the same secret key with

$$\hat{ct}[SK] = ct^*[SK] - \sum_i c_{3,i}^* e_i.$$

The $EK^{(i)}$ are published with the public key as the **evaluation key**. BFV, Section 9 and BGV, Section 8 produce the three-component ct^* ; BFV, Section 10, BGV, Section 9 and CKKS, Section 5 then relinearise exactly as above, and all three cite Lecture 4, Section 4 for the reading $ct = c_0 + c_1 S$ rather than redo it. Nothing here mentions a plaintext modulus or a scale, which is why the three schemes can share it.

Example 4 (A relinearisation in full). Take $q = 64$, $B = 4$, $\ell = 3$ and $SK = 3$, so $SK^2 = 9$. We work with $n = 1$, i.e. $R_q = \mathbb{Z}_{64}$, so that every product is an integer product. Let the degree-2 ciphertext be

$$ct^* = (c_1^*, c_2^*, c_3^*) = (5, 12, 45), \quad ct^*[SK] = 5 + 12 \cdot 3 + 45 \cdot 9 = 446 \equiv -2 \pmod{64}.$$

Its top component is the $a = 45$ of Section 3, so $G^{-1}(45) = (1, -1, -1)$. Sample $a_i = (7, 23, 51)$ and $e_i = (1, -1, 1)$; then

$$\begin{aligned} EK_1^{(0)} &= -(7 \cdot 3 + 1) + 1 \cdot 9 = -13, \\ EK_1^{(1)} &= -(23 \cdot 3 - 1) + 4 \cdot 9 = -32, & EK_1^{(i)} + a_i SK &= B^i SK^2 - e_i \checkmark \\ EK_1^{(2)} &= -(51 \cdot 3 + 1) + 16 \cdot 9 = -10, \end{aligned}$$

(for instance $-13 + 21 = 8 = 9 - 1$). Contracting against the digits,

$$\hat{c}_1 = 5 + [(-13)(1) + (-32)(-1) + (-10)(-1)] = 5 + 29 = 34, \quad \hat{c}_2 = 12 + [7 - 23 - 51] = -55 \equiv 9,$$

and $\hat{ct}[SK] = 34 + 9 \cdot 3 = 61 \equiv -3 \pmod{64}$. The predicted answer is $ct^*[SK] - \sum_i c_{3,i}^* e_i = -2 - (1 + 1 - 1) = -3$. $\checkmark$ The added error is 1.

5. Batching: SIMD Slots

Theorem 1 (Batching). *Let t be prime with $t \equiv 1 \pmod{2n}$ and let $\zeta \in \mathbb{F}_t^\times$ have order $2n$. Then*

$$R_t = \mathbb{F}_t[X]/(X^n + 1) \cong \prod_{j \in \mathbb{Z}_{2n}^\times} \mathbb{F}_t[X]/(X - \zeta^j) \cong \mathbb{F}_t^n,$$

*the isomorphism being $a \mapsto (a(\zeta^j))_{j \in \mathbb{Z}_{2n}^\times}$, and it respects $+$ and $\times$ componentwise. Its n components are the **slots**, with $\text{Decode}(a) = (a(\zeta), a(\zeta^3), \dots, a(\zeta^{2n-1}))$ and $\text{Encode} = \text{Decode}^{-1}$.*

Because BGV's and BFV's operations act on R_t , one homomorphic addition performs n independent additions and one homomorphic multiplication n independent multiplications: this is **batching**, and it is what makes the amortised cost per data item tolerable. BFV, Section 5 selects the plaintext modulus by this congruence for exactly that reason, GSW, Section 1 cites it as the packing GSW gives up, and CKKS, Section 3 is the same construction over $\mathbb{C}$ (Section 7). Encode is the inverse negacyclic NTT of Lecture 3, so it is free.

Example 5 ($n = 4$, $t = 17$: four slots). Here $17 \equiv 1 \pmod{8}$. From Lecture 2, 3 is a primitive root of $\mathbb{F}_{17}$, so $\zeta = 3^2 = 9$ has order 8 with powers $(9^0, \dots, 9^7) = (1, 9, 13, 15, 16, 8, 4, 2)$. The four primitive 8-th roots — the odd powers — are

$$\zeta^1 = 9, \quad \zeta^3 = 15, \quad \zeta^5 = 8, \quad \zeta^7 = 2,$$

each satisfying $x^4 = 16 = -1$; and indeed $(X - 9)(X - 15)(X - 8)(X - 2) = X^4 + 1$ over $\mathbb{F}_{17}$, since the roots sum to $34 \equiv 0$ and multiply to $2160 \equiv 1$.

Encode. For the slot vector $z = (3, 1, 4, 1)$, interpolating through the nodes $(9, 15, 8, 2)$ gives $\mu(X) = 15 + 8X + 5X^2 + 15X^3$.

Decode. Using $9^2 = 13$, $9^3 = 15$ and similarly for the others,

$$\begin{aligned}\mu(9) &= 15 + 72 + 65 + 225 = 377 \equiv 3, \\ \mu(15) &= 15 + 120 + 20 + 135 = 290 \equiv 1, \\ \mu(8) &= 15 + 64 + 65 + 30 = 174 \equiv 4, \\ \mu(2) &= 15 + 16 + 20 + 120 = 171 \equiv 1,\end{aligned} \quad \implies \quad (3, 1, 4, 1). \checkmark$$

One multiplication, four products. Encode $z' = (2, 0, 5, 7)$, obtaining $\mu'(X) = 12 + 10X + 16X^3$. Their product in R_{17} , by the negacyclic rule $X^4 = -1$, is $\mu\mu' = 4 + 13X + 2X^2 + 11X^3$, whose slots are

$$(6, 0, 3, 7) = (3 \cdot 2, 1 \cdot 0, 4 \cdot 5, 1 \cdot 7) \bmod 17. \checkmark$$

6. Galois Automorphisms and Rotation

Definition 2 (Galois automorphisms, and the slot permutation). For odd k , let $\sigma_k : R \rightarrow R$ be $a(X) \mapsto a(X^k) \bmod (X^n + 1)$. Each σ_k is a ring automorphism, $\sigma_k = \sigma_{k'}$ exactly when $k \equiv k' \pmod{2n}$, and they form the group $\mathbb{Z}_{2n}^\times = \langle 5 \rangle \times \langle -1 \rangle$ with 5 of order $n/2$. On the slots of Theorem 1,

$$(\sigma_k(a))(\zeta^j) = a(\zeta^{jk}),$$

so σ_k permutes slots by $j \mapsto jk$ in $\mathbb{Z}_{2n}^\times$.

This is how data moves between slots, which $+$ and $\times$ cannot do. Applied to a ciphertext, σ_k is an index permutation with sign flips and costs no multiplications, but the result decrypts under $\sigma_k(SK)$ rather than SK , so every rotation is followed by a **key switch**: Proposition 3 with SK^2 replaced by $\sigma_k(SK)$, and one switching key published per rotation amount used. CKKS, Section 3 rotates its slots with σ_5 and conjugates them with σ_{-1} ; BGV, Section 9 notes that rotation and relinearisation are the same machinery; and CKKS bootstrapping is built almost entirely from sums of rotations.

Example 6 (σ_5 acting on slots, $n = 4$, $t = 17$). First the group: $\mathbb{Z}_8^\times = \{1, 3, 5, 7\}$ and every element squares to 1 ($3^2 = 9 \equiv 1$, $5^2 = 25 \equiv 1$, $7^2 = 49 \equiv 1$), so it is the Klein four-group; $\langle 5 \rangle = \{1, 5\}$ has order $2 = n/2$, $\langle -1 \rangle = \{1, 7\}$, and $5 \cdot 7 = 35 \equiv 3$, so the two subgroups generate everything. $\checkmark$

Now take $a(X) = 1 + 2X + 3X^2 + 4X^3$, the running example of Lecture 2. Its slots at $(\zeta, \zeta^3, \zeta^5, \zeta^7) = (9, 15, 8, 2)$ are

$$(a(9), a(15), a(8), a(2)) = (16, 11, 13, 15).$$

Apply σ_5 . Since $X^5 = -X$, $X^{10} = X^2$ and $X^{15} = -X^3$,

$$\sigma_5(a) = 1 - 2X + 3X^2 - 4X^3 = 1 + 15X + 3X^2 + 13X^3 \quad \text{in } R_{17},$$

whose slots are $(13, 15, 16, 11)$. Compare with the original: the map $j \mapsto 5j$ on $\mathbb{Z}_8^\times$ sends $1 \mapsto 5$, $3 \mapsto 7$, $5 \mapsto 1$, $7 \mapsto 3$, i.e. the permutation $(1\ 5)(3\ 7)$, and that is exactly the rearrangement observed. $\checkmark$

7. The Canonical Embedding and CKKS Encoding

Definition 3 (Canonical embedding, and CKKS encoding). Let $K = \mathbb{Q}[X]/(X^n + 1)$ and $\zeta = \exp(\pi i/n)$. The **canonical embedding** is

$$\sigma : K \longrightarrow \mathbb{C}^n, \quad \sigma(a) = (a(\zeta), a(\zeta^3), \dots, a(\zeta^{2n-1})),$$

an injective ring homomorphism. Keeping one coordinate from each conjugate pair, namely the subgroup $\langle 5 \rangle \subset \mathbb{Z}_{2n}^\times$, gives the $\mathbb{R}$ -linear multiplicative bijection $\tau : K \rightarrow \mathbb{C}^{n/2}$, $\tau(a) = (a(\zeta), a(\zeta^5), a(\zeta^{25}), \dots)$, and for a scale factor Δ ,

$$\text{Encode}(z, \Delta) = \lfloor \Delta \cdot \tau^{-1}(z) \rfloor \in R, \quad \text{Decode}(\mu, \Delta) = \Delta^{-1} \tau(\mu) \in \mathbb{C}^{n/2}.$$

CKKS, Section 3 takes both maps over unchanged and cites Lecture 4, Section 7 for them. The $n/2$ coordinates of $\tau(a)$ are the CKKS slots, the complex analogue of Section 5's, and because τ is multiplicative, ring multiplication is slotwise multiplication; the choice of $\langle 5 \rangle$ is what makes σ_5 of Section 6 a cyclic rotation of them and σ_{-1} slotwise conjugation. Lecture 5, Section 8 reuses σ for a different purpose: $\sigma(R)$ is a lattice, and Ring-LWE hardness is stated in its geometry.

Example 7 ($n = 4$, $\Delta = 10^2$). Here τ maps into $\mathbb{C}^2$ and $\zeta = \exp(\pi i/4) = \frac{\sqrt{2}}{2}(1 + i)$. Let $z = (1 + 4i, 5 - 2i)$.

Encode. The evaluation matrix $A_{k,j} = \zeta^{kj}$ for $k \in \{1, 3, 5, 7\}$ satisfies $A^*A = 4I$, so τ^{-1} is the conjugate transpose divided by 4: writing $v_1 = \Delta z_1$, $v_5 = \Delta z_2$, $v_7 = \overline{v_1}$, $v_3 = \overline{v_5}$, we have $(\Delta \tau^{-1}(z))_j = \frac{1}{4} \sum_k \zeta^{-kj} v_k$. With $v_1 = 100 + 400i$, $v_3 = 500 + 200i$, $v_5 = 500 - 200i$, $v_7 = 100 - 400i$ this gives

$$\Delta \cdot \tau^{-1}(z) = (300, 50\sqrt{2}, 100, 250\sqrt{2}) = (300, 70.7107, 100, 353.5534),$$

all four entries real, as they must be. Rounding, $\mu(X) = 300 + 71X + 100X^2 + 354X^3$.

Decode. Using $\zeta^2 = i$, $\zeta^5 = -\zeta$ and $\zeta^{15} = \zeta^7 = \bar{\zeta}$,

$$\begin{aligned} \mu(\zeta) &= 300 + 50.2046(1 + i) + 100i + 250.3158(-1 + i) = 99.8888 + 400.5204i, \\ \mu(\zeta^5) &= 300 - 50.2046(1 + i) + 100i + 250.3158(1 - i) = 500.1112 - 200.5204i, \end{aligned}$$

so $\text{Decode}(\mu, \Delta) = (0.99889 + 4.00520i, 5.00111 - 2.00520i)$, against the true z . Each slot is off by 5.3×10^{-3} .

Summary

1. RNS is CRT coefficientwise: $R_Q \cong \prod_i R_{q_i}$, with the chain $q_\ell = p_0 \cdots p_\ell$ and level ℓ . Dropping a level deletes one component.
2. Modulus switching sends $ct \mapsto \lfloor (q'/q) ct \rfloor$ in the class modulo t , preserving the message and dividing the noise by q/q' , subject to $q \equiv q' \pmod{t}$. CKKS's RESCALE is the same formula without the congruence.
3. Base- B decomposition gives $\langle g, G^{-1}(a) \rangle = a$ with $\|G^{-1}(a)\|_\infty \leq B/2$ regardless of a . At $q = 64$, $B = 4$, $G^{-1}(45) = (1, -1, -1)$.
4. Read ct as $c_0 + c_1 S \in R_q[S]$; multiplication gives a quadratic in S , and relinearisation contracts $G^{-1}(c_3^*)$ against an evaluation key encrypting $g \cdot SK^2$, at the cost of $\sum_i c_{3,i}^* e_i$. Common to BFV, BGV and CKKS.
5. When $t \equiv 1 \pmod{2n}$, $R_t \cong \mathbb{F}_t^n$: n SIMD slots with componentwise $+$ and $\times$, computed by the NTT.
6. The maps $\sigma_k : X \mapsto X^k$, k odd, form $\mathbb{Z}_{2n}^\times = \langle 5 \rangle \times \langle -1 \rangle$ and permute slots by $j \mapsto jk$; each must be followed by a key switch, item 4 with SK^2 replaced by $\sigma_k(SK)$.

7. The canonical embedding evaluates at the complex roots of $X^n + 1$ and reduces to $\tau : K \rightarrow \mathbb{C}^{n/2}$. CKKS encodes by $\lfloor \Delta \tau^{-1}(z) \rfloor$ and decodes by $\Delta^{-1} \tau(\cdot)$.

Exercises

1. Recover the integer with RNS representation $(3, 4, 26)$ modulo $Q = 67609$.
2. Redo the worked switch of Section 2 with $m = 3$, $v = 2$ and $c_1 = 12345$.
3. Decompose $a = 45 \in \mathbb{Z}_{64}$ in base $B = 2$ and verify $\langle g, G^{-1}(a) \rangle = a$.
4. Redo the relinearisation of Section 4 with the errors $e_i = (-1, 1, -1)$.
5. Compute the slots of $b = 4 + 3X + 2X^2 + X^3$ at $n = 4$, $t = 17$, and the slots of $\sigma_5(b)$.